BACKTRACK: measurement and evaluation

The problem is a learner who cannot complete today's task and does not know what to review. The evaluation asks whether BACKTRACK's targeted route helps them return to that task and retain the skill.

Primary question

Does a destination-focused BACKTRACK plus Khan workflow produce more retained reentry than a realistic teacher-selected Khan routine with comparable access and support?

Signature outcome: retained reentry

For each learner, prespecify one eligible current destination. A baseline-blocked learner scores below the approved criterion on the supervised baseline. Documented retained reentry requires passing both a fresh immediate post-route assessment and a parallel assessment 7-14 days later, without answer assistance.

Proposed scoring: two destination problems, five rubric points each, passing at least 8/10 with the critical target step correct on both problems. Educators must approve the rubric, difficulty, and threshold before enrollment. This is an assessment design proposal, not a validated psychometric scale. A correct numerical answer without the required reasoning may not demonstrate the intended skill.

The primary reported proportion is:

Learners meeting both post and delayed criteria / all eligible baseline-blocked learners enrolled in the evaluated cohort.

Missing post or delayed data counts as no documented retained reentry in that primary summary. Also report measured failure, absence, withdrawal, and unusable data separately. Provide an observed-case sensitivity result and the number behind it. Never relabel missing learners as tested failures or silently drop them.

Secondary outcomes

- Immediate reentry: passes the post criterion, with the same enrolled denominator.
- Active time to first reentry: supervised active task, assessment, and practice minutes until the first qualifying post task. Record elapsed calendar time separately.

- Repeated relevant Khan use: at least one matched exercise assignment completed in at least three of four offered teaching weeks. Report all assigned learners as the denominator, and report the offered weeks.
- Delayed score and score change: retain continuous scores alongside thresholds.
- Teacher burden: setup, assigning, troubleshooting, reporting, and support minutes, separated by initial and repeated use.
- Learner experience: completion, voluntary continuation, frustration/pressure, and willingness to return.

The repeated-use definition is a planning operationalization, not Khan's official definition of meaningful use. Adjust it before the pilot if the reporting source cannot support it, and record the change.

Time-to-reentry without survivor bias

Start the clock at the first post-baseline intervention activity. Count observed active minutes consistently in both conditions. Do not infer learning time from a browser tab being open. Pause active timing during outages and record the outage separately. For self-study use without supervision, label time as self-reported or unavailable.

Learners who have not reentered remain in the analysis as censored at their last valid observation or the prespecified study end. Present the cumulative share reentering by common exposure points when sample size permits. Report a median only if at least half reenter and the data supports it. A mean among successful finishers may be descriptive, but cannot be the headline efficiency claim.

Formal and public evidence

The formal cohort has permissions, baseline eligibility, supervised tasks, a participant key, attendance, and source reconciliation. Public visitors have none of those assurances. Their device-local answers are product evidence only. Do not add public visitors to the school cohort, count page views as beneficiaries, or label an unsupervised correct answer a documented school recovery.

The current product requires two distinct unassisted checks before removing a skill review. That is a conservative product rule relative to the inherited one-answer rule, not proof of mastery. No calibrated confidence interval for a learner's knowledge is claimed.

Comparison design

Preferred design, subject to school approval: randomize eligible learners within school and destination, stratifying on baseline performance where feasible, to immediate BACKTRACK-supported Khan practice or equal-time teacher-selected Khan practice. Preserve normal teaching and access to adult help. Use a delayed start so the comparison group receives the route after its outcome window.

Assign once, document allocation before intervention, and analyze by original assignment. Assessors should not know the learner's condition where practical. Record contamination, additional tutoring, AI use, and protocol departures. Use the same access windows and parallel assessment policy in both groups.

If individual allocation disrupts teaching, consider group assignment with enough groups for inference. With only two or three clusters, do not claim a precise cluster-randomized effect. If comparison design fails, report pre/post feasibility outcomes and explicitly state that normal instruction, selection, practice, or regression to the mean could explain improvement.

Sample size and analysis

The 80-120 learner target is a feasibility target, not a power calculation. Before a confirmatory trial, a qualified evaluator should calculate required sample size using a justified baseline rate, smallest meaningful difference, clustering, and likely attrition. Do not choose an effect size because it fits the available grant.

Report group counts, baseline distributions, missingness, proportions, and uncertainty intervals. For causal estimates, use a prespecified approach appropriate to allocation and clustering. Separate exploratory destination-level findings from the primary learner-level outcome. Do not promote an attractive subgroup after seeing results.

Baseline, post, and delayed task construction

Create forms A, B, and C that test the same target operation with different numbers and surface features. Review for difficulty and alternate valid methods. Counterbalance forms where possible. Keep the formal forms outside public practice, and do not show their solutions before all relevant checks are complete. Double-score a subset and resolve rubric disagreement.

Evidence dictionary

- participant_id: random school-issued pseudonym; identifying key stays at school.
- cohort_id, educator_id, destination_id, route_version, assigned_condition.
- permission_status, eligibility_status, baseline_date, baseline_score, baseline_form.
- session_id, date, offered_minutes, attended_minutes, active_minutes, access_failure.
- khan_assignment_id, mapped_skill, report_date, reported_completion, source_type.
- post_date, post_score, post_form, assistance_flag, assessor_id.
- delayed_due_date, delayed_date, delayed_score, delayed_form, missing_reason.
- teacher_setup_minutes, teacher_repeat_minutes, facilitator_minutes, support_reason.
- direct_cash_cost, in_kind_hours, consented_feedback, protocol_departure.

Do not collect unnecessary personal narratives or identifiable screenshots. The pseudonym alone does not make a dataset anonymous if the school can reconnect it to a learner.

Route efficiency and retention tests

Count reviews actually removed from a declared provisional route. Do not turn this into minutes saved until observed comparable durations support the estimate. Evaluate optional continuation at a genuine stopping point, reporting the number offered, the number choosing it, and the number completing it. Compare a plain continuation invitation with the route-benefit invitation only under an approved, prespecified test. Measure pressure alongside uptake.

Research basis and limits

IES guidance and retrieval/spacing research support including later checks and worked examples. They do not establish BACKTRACK's effectiveness.[1][2] Foster's confidence-assessment trial did not find improved attainment; its negative-marking intervention differs from BACKTRACK's non-graded metadata. We must not cite it as proof that confidence-aware routing improves scores.[3]

Sources

[1] IES, Organizing Instruction and Study to Improve Student Learning, 2007. https://ies.ed.gov/ncee/wwc/practiceguide/1

[2] Carpenter, Pan & Butler, The science of effective learning with spacing and retrieval practice, 2022. https://doi.org/10.1038/s44159-022-00089-1

[3] Foster, Implementing Confidence Assessment in Low-Stakes, Formative Mathematics Assessments, published 2021. https://doi.org/10.1007/s10763-021-10207-9